Class 12 Mathematics

Chapter 8: Application of Integrals

School Worksheet: step-wise solutions

20 questions **53** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 MCQ — 1 mark

The area of the region bounded by $y = x^2$, the x-axis and the lines $x = 0$ and $x = 3$ is:

(A) 3 sq units **(B)** 9 **sq units**

(C) 27 sq units **(D)** 6 sq units

Formula / idea: Area under $y = f(x) \geq 0$ from $x = a$ to $x = b$ is $\int_a^b y\,dx$.

1 Write the area as an integral.

$$A = \int_0^3 x^2\,dx$$

2 Integrate and put the limits. ✓ 1

$$A = \left[\frac{x^3}{3}\right]_0^3 = \frac{27}{3} - 0 = 9$$

3 So the area is 9 sq units, option (B).

Answer **(B)** 9 **sq units**

Watch out: Forgetting to divide by 3 and choosing 27.

2 MCQ — 1 mark

The area enclosed by the circle $x^2 + y^2 = 4$ is:

(A) 2π sq units **(B)** 4π **sq units**

(C) 16π sq units **(D)** π sq units

Formula / idea: Area of the circle $x^2 + y^2 = r^2$ is πr^2.

1 Compare with $x^2 + y^2 = r^2$.

$r^2 = 4 \Rightarrow r = 2$

2 Use area $= \pi r^2$. ✓ 1
$A = \pi(2)^2 = 4\pi$
Same as $4\int_0^2 \sqrt{4 - x^2}\, dx$.

3 So the area is 4π sq units, option (B).

Answer **(B) 4π sq units**

Watch out: Using $r = 4$ instead of $r = 2$ and choosing 16π.

3 MCQ — 1 mark

The area enclosed by the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ is:

(A) 6π **sq units** **(B)** 36π sq units
(C) 13π sq units **(D)** 12π sq units

Formula / idea: Area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is πab.

1 Compare with $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.
$a^2 = 9,\ b^2 = 4 \Rightarrow a = 3,\ b = 2$

2 Use area $= \pi ab$. ✓ 1
$A = \pi(3)(2) = 6\pi$

3 So the area is 6π sq units, option (A).

Answer **(A) 6π sq units**

Watch out: Multiplying a^2 and b^2 to get 36π.

4 MCQ — 1 mark

The area of the region bounded by $y = \cos x$, the x-axis, $x = 0$ and $x = \frac{\pi}{2}$ is:

(A) 2 sq units **(B)** 1 **sq unit** **(C)** $\frac{\pi}{2}$ sq units **(D)** 0

Formula / idea: $\int \cos x\, dx = \sin x + C$

1 Write the area as an integral.
$A = \int_0^{\pi/2} \cos x\, dx$
$\cos x \geq 0$ on $\left[0, \frac{\pi}{2}\right]$.

2 Integrate and put the limits. ✓ 1

$$A = [\sin x]_0^{\pi/2} = \sin\frac{\pi}{2} - \sin 0 = 1 - 0 = 1$$

3 So the area is 1 sq unit, option (B).

Answer **(B)** 1 **sq unit**

5 MCQ 1 mark

The area of the region bounded by the curve $y = x$ and the x-axis, between $x = 0$ and $x = 2$, is:

(A) 2 **sq units** **(B)** $\frac{1}{2}$ sq unit **(C)** 1 sq unit **(D)** 4 sq units

1 Write the area as an integral.

$$A = \int_0^2 x\,dx$$

2 Integrate and put the limits. ✓ 1

$$A = \left[\frac{x^2}{2}\right]_0^2 = \frac{4}{2} - 0 = 2$$

Check: triangle area $\frac{1}{2} \times 2 \times 2 = 2$.

3 So the area is 2 sq units, option (A).

Answer **(A)** 2 **sq units**

6 **Assertion-Reason** 1 mark

Assertion (A): The area bounded by $y = \sin x$ and the x-axis between $x = 0$ and $x = 2\pi$ is 4 sq units.

Reason (R): $\int_0^{2\pi} \sin x \, dx = 0$.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) **Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).**

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: Area uses $|\sin x|$: split where the curve crosses the x-axis.

1 Assertion: $\sin x < 0$ on $(\pi, 2\pi)$, so split at π.

$$A = \int_0^{\pi} \sin x \, dx + \left| \int_{\pi}^{2\pi} \sin x \, dx \right|$$
$$= 2 + |-2| = 4$$

2 So **A is true**.

3 Reason: integrate straight from 0 to 2π.

$$\int_0^{2\pi} \sin x \, dx = [-\cos x]_0^{2\pi} = -1 + 1 = 0$$

4 So **R is true**.

5 R gives 0 because the parts cancel; it does not give the area 4. ✓ 1

6 So R is not the correct explanation of A: option (B).

Answer **(B) Both A and R are true, but R is not the correct explanation of A**

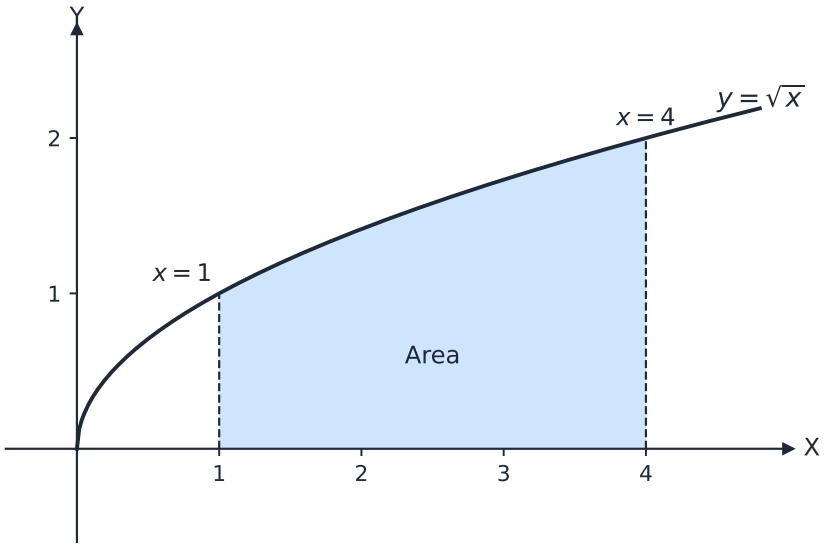
Watch out: Thinking area $= \int_0^{2\pi} \sin x \, dx = 0$ and marking A false.

7 **Short answer** 2 marks

Find the area of the region bounded by $y^2 = x$, the x-axis and the lines $x = 1$ and $x = 4$, lying in the first quadrant.

Formula / idea: Area $= \int_a^b y\,dx$, with y taken from the curve.

1 Draw a rough sketch and shade the region from $x = 1$ to $x = 4$.



2 In the first quadrant, take the positive root.

$y^2 = x \Rightarrow y = \sqrt{x}$

3 Write the required area. ✓ 1

$$A = \int_1^4 y\,dx = \int_1^4 \sqrt{x}\,dx$$

4 Integrate. ✓ ½

$$A = \left[\frac{2}{3}x^{3/2}\right]_1^4$$

5 Put the limits.

$$A = \frac{2}{3}\left(4^{3/2} - 1^{3/2}\right) = \frac{2}{3}(8 - 1) = \frac{14}{3}$$

$4^{3/2} = (\sqrt{4})^3 = 8$

6 Hence the required area $= \frac{14}{3}$ sq units. ✓ ½

Answer $\frac{14}{3}$ **sq units**

Watch out: Taking $4^{3/2}$ as 6 instead of 8.

Level 2

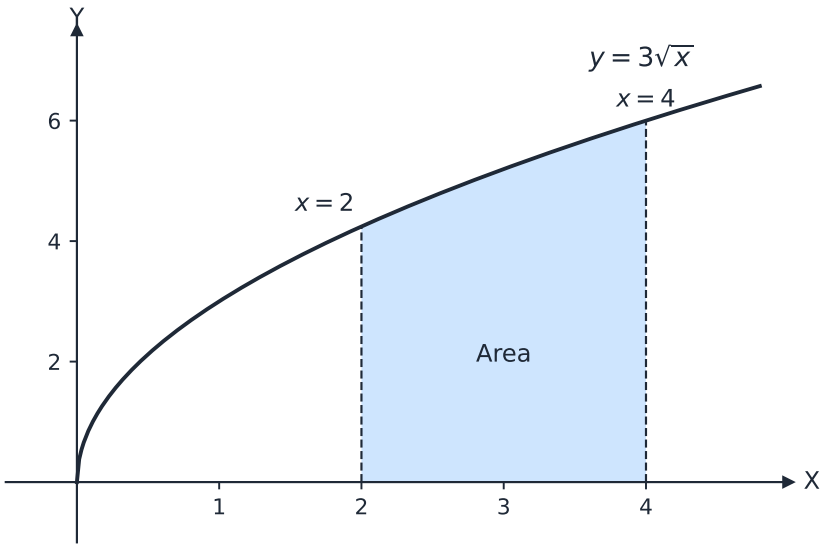
18 marks

8 **Short answer** 3 marks

Find the area of the region bounded by $y^2 = 9x$, $x = 2$, $x = 4$ and the x-axis in the first quadrant.

Formula / idea: Area $= \int_a^b y\,dx$, with y taken from the curve.

1 Draw a rough sketch and shade the region from $x = 2$ to $x = 4$. ✓ ½



2 In the first quadrant, take the positive root.
$y^2 = 9x \Rightarrow y = 3\sqrt{x}$

3 Write the required area. ✓ 1
$$A = \int_2^4 y\,dx = \int_2^4 3\sqrt{x}\,dx$$

4 Integrate. ✓ ½
$$A = 3\left[\frac{2}{3}x^{3/2}\right]_2^4 = 2\left[x^{3/2}\right]_2^4$$

5 Put the limits. ✓ ½
$$A = 2\left(4^{3/2} - 2^{3/2}\right) = 2\left(8 - 2\sqrt{2}\right)$$
$2^{3/2} = 2\sqrt{2}$

6 Hence the required area $= 16 - 4\sqrt{2}$ sq units. ✓ ½

Answer $16 - 4\sqrt{2}$ **sq units**

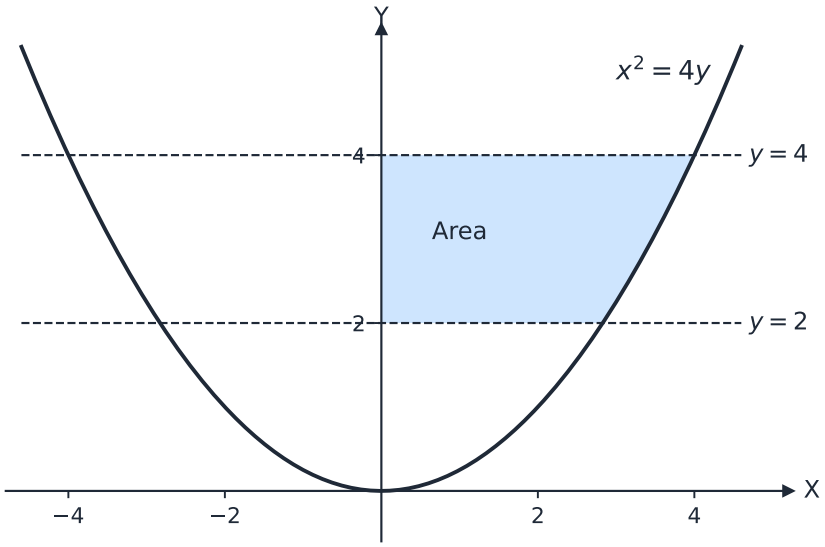
Watch out: Writing $2^{3/2}$ as 3 or as $\sqrt{2}$; it is $2\sqrt{2}$.

9 **Short answer** 3 marks

Find the area of the region bounded by $x^2 = 4y$, $y = 2$, $y = 4$ and the y-axis in the first quadrant.

Formula / idea: Area between a curve and the y-axis from $y = c$ to $y = d$ is $\int_c^d x\,dy$.

1 Draw a rough sketch and shade the region between the y-axis and the curve. ✓ ½



2 The region is bounded by the y-axis, so integrate w.r.t. y.

3 In the first quadrant, express x in terms of y.

$x^2 = 4y \Rightarrow x = 2\sqrt{y}$

4 Write the required area. ✓ 1

$$A = \int_2^4 x\,dy = \int_2^4 2\sqrt{y}\,dy$$

5 Integrate. ✓ ½

$$A = 2\left[\frac{2}{3}y^{3/2}\right]_2^4 = \frac{4}{3}\left[y^{3/2}\right]_2^4$$

6 Put the limits. ✓ ½

$$A = \frac{4}{3}\left(8 - 2\sqrt{2}\right)$$

7 Hence the required area $= \dfrac{32 - 8\sqrt{2}}{3}$ sq units. ✓ ½

Answer $\dfrac{32 - 8\sqrt{2}}{3}$ **sq units**

Watch out: Integrating w.r.t. x; here the strips are horizontal, so use dy.

10 **Short answer** 3 marks

Using integration, find the area of the region bounded by the line $y = 3x + 2$, the x-axis and the ordinates $x = -1$ and $x = 1$.

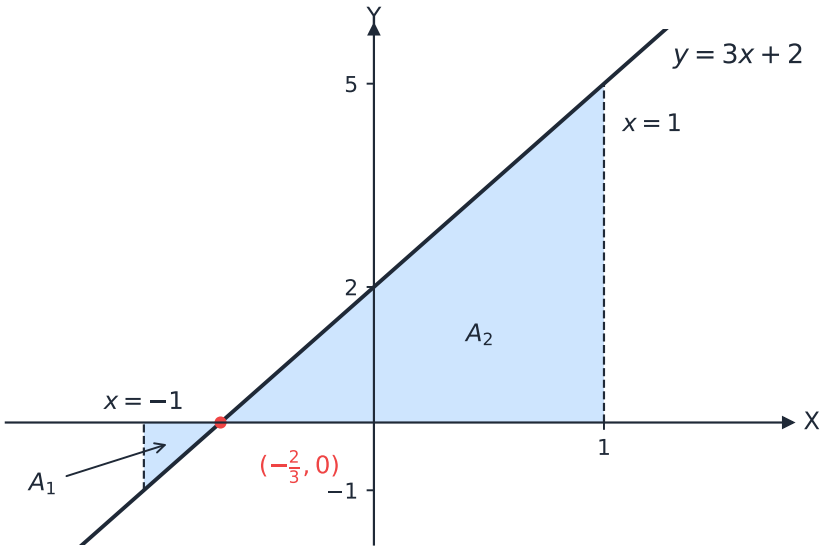
Formula / idea: A part of the region below the x-axis is taken with modulus.

1 Find where the line meets the x-axis.

$3x + 2 = 0 \Rightarrow x = -\dfrac{2}{3}$

Line is below the x-axis for $x < -\frac{2}{3}$.

2 Draw a rough sketch; the region has two parts A_1 and A_2. ✓ ½



3 Split at $x = -\dfrac{2}{3}$ and write the area. ✓ 1

$$A = \left| \int_{-1}^{-2/3} (3x + 2)\, dx \right| + \int_{-2/3}^{1} (3x + 2)\, dx$$

4 Evaluate the first integral. ✓ ½

$$\left[\frac{3x^2}{2} + 2x\right]_{-1}^{-2/3} = \left(\frac{2}{3} - \frac{4}{3}\right) - \left(\frac{3}{2} - 2\right)$$
$$= -\frac{2}{3} + \frac{1}{2} = -\frac{1}{6}$$

5 Evaluate the second integral. ✓ ½

$$\left[\frac{3x^2}{2} + 2x\right]_{-2/3}^{1} = \frac{7}{2} - \left(-\frac{2}{3}\right) = \frac{25}{6}$$

6 Add the two areas.

$$A = \left|-\frac{1}{6}\right| + \frac{25}{6} = \frac{26}{6} = \frac{13}{3}$$

7 Hence the required area $= \dfrac{13}{3}$ sq units. ✓ ½

Answer $\dfrac{13}{3}$ **sq units**

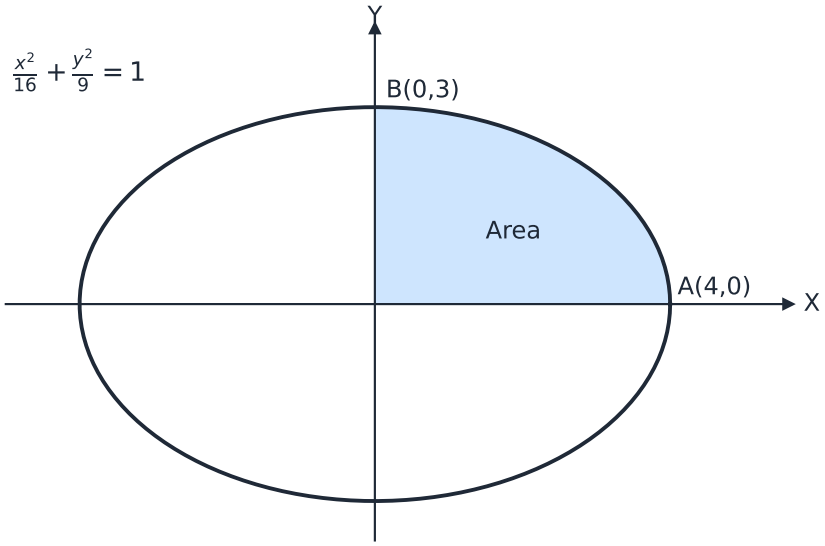
Watch out: Integrating straight from -1 to 1 gives 4, which is wrong.

11 **Short answer** 2 marks

Find the area of the region in the first quadrant bounded by the ellipse $\dfrac{x^2}{16} + \dfrac{y^2}{9} = 1$ and the coordinate axes.

Formula / idea: $\int \sqrt{a^2 - x^2}\,dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C$

1 Draw a rough sketch; shade the first-quadrant part of the ellipse.



2 Solve for y in the first quadrant.

$$y^2 = \frac{9}{16}\left(16 - x^2\right) \Rightarrow y = \frac{3}{4}\sqrt{16 - x^2}$$

3 Write the required area. ✓ 1

$$A = \int_0^4 y\,dx = \frac{3}{4}\int_0^4 \sqrt{16 - x^2}\,dx$$

4 Integrate using the $\sqrt{a^2 - x^2}$ formula with $a = 4$. ✓ ½

$$A = \frac{3}{4}\left[\frac{x}{2}\sqrt{16 - x^2} + 8\sin^{-1}\frac{x}{4}\right]_0^4$$

5 Put the limits.

$$A = \frac{3}{4}\left[\left(0 + 8\cdot\frac{\pi}{2}\right) - 0\right] = \frac{3}{4} \times 4\pi$$

$\sin^{-1} 1 = \frac{\pi}{2}$, $\sin^{-1} 0 = 0$

6 Hence the required area $= 3\pi$ sq units. ✓ ½

Answer 3π **sq units**

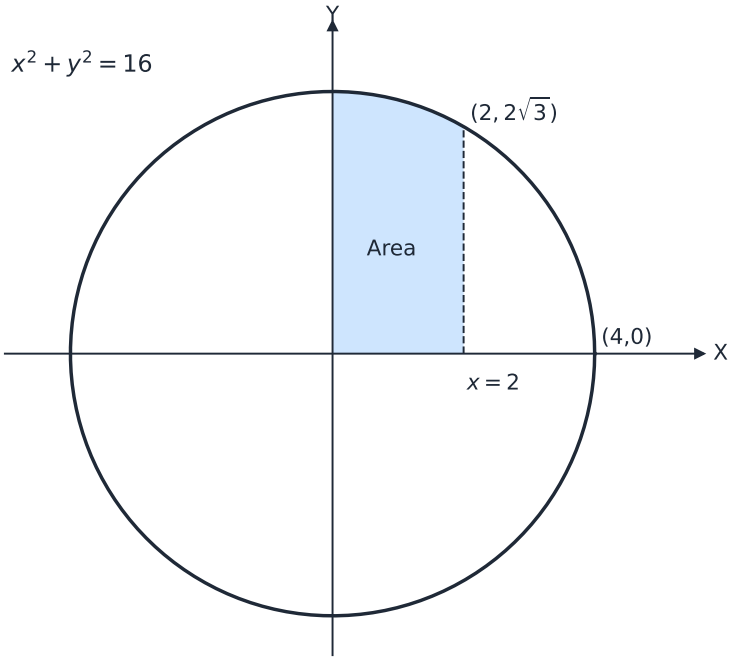
Watch out: Writing $\frac{a^2}{2} = 16$ instead of 8 in the formula.

12 **Short answer** 3 marks

Find the area of the region bounded by the circle $x^2 + y^2 = 16$, the x-axis and the lines $x = 0$ and $x = 2$, lying in the first quadrant.

Formula / idea: $\int \sqrt{a^2 - x^2}\,dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C$

1 Draw a rough sketch; shade the region from $x = 0$ to $x = 2$. ✓ ½



2 Solve for y in the first quadrant.

$y = \sqrt{16 - x^2}$

3 Write the required area. ✓ 1

$$A = \int_0^2 \sqrt{16 - x^2}\,dx$$

4 Integrate using the $\sqrt{a^2 - x^2}$ formula with $a = 4$. ✓ ½

$$A = \left[\frac{x}{2}\sqrt{16 - x^2} + 8\sin^{-1}\frac{x}{4}\right]_0^2$$

5 Put the limits. ✓ ½

$$A = \left(\frac{2}{2}\sqrt{12} + 8\sin^{-1}\frac{1}{2}\right) - 0$$

$$= 2\sqrt{3} + 8 \cdot \frac{\pi}{6}$$

$\sin^{-1}\frac{1}{2} = \frac{\pi}{6}$

6 Hence the required area $= 2\sqrt{3} + \dfrac{4\pi}{3}$ sq units. ✓ ½

Answer $2\sqrt{3} + \dfrac{4\pi}{3}$ **sq units**

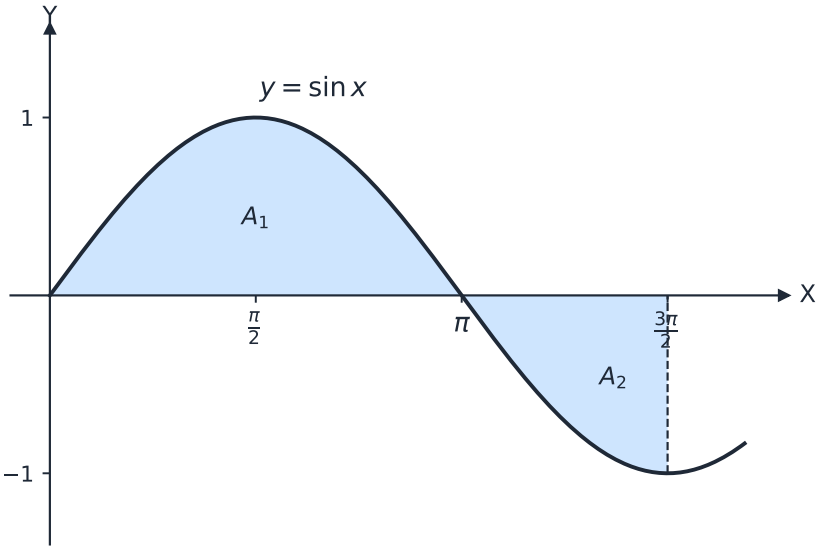
Watch out: Taking $\sin^{-1}\frac{1}{2} = \frac{\pi}{3}$ instead of $\frac{\pi}{6}$.

13 **Short answer** 2 marks

Find the area bounded by $y = \sin x$ and the x-axis between $x = 0$ and $x = \dfrac{3\pi}{2}$.

Formula / idea: A part of the region below the x-axis is taken with modulus.

1 Sketch: $\sin x \geq 0$ on $[0, \pi]$ and $\sin x \leq 0$ on $\left[\pi, \dfrac{3\pi}{2}\right]$.



2 Split at $x = \pi$ and write the area. ✓ 1

$$A = \int_0^{\pi} \sin x \, dx + \left| \int_{\pi}^{3\pi/2} \sin x \, dx \right|$$

3 Evaluate each integral. ✓ ½

$$\int_0^{\pi} \sin x \, dx = [-\cos x]_0^{\pi} = 1 + 1 = 2$$

$$\int_{\pi}^{3\pi/2} \sin x \, dx = [-\cos x]_{\pi}^{3\pi/2} = 0 - 1 = -1$$

4 Add the two areas.

$$A = 2 + |-1| = 3$$

5 Hence the required area $= 3$ sq units. ✓ ½

Answer 3 **sq units**

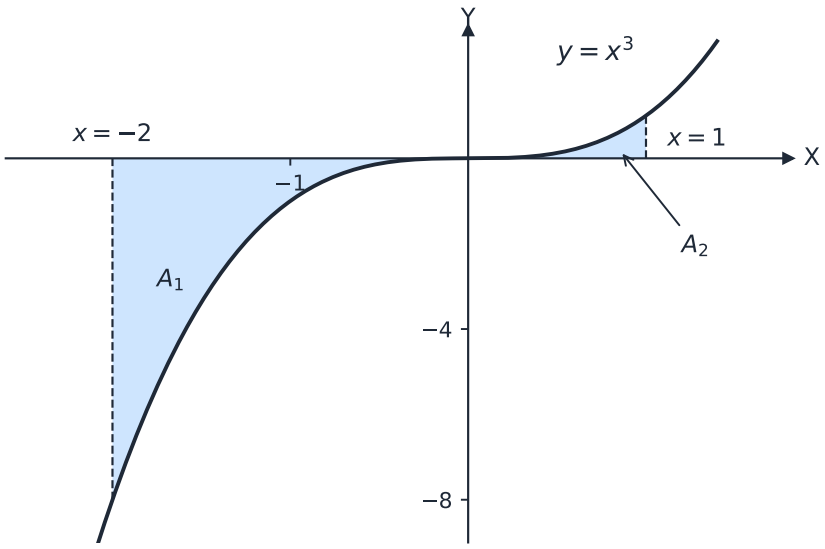
Watch out: Integrating straight from 0 to $\frac{3\pi}{2}$ gives 1, not the area.

14 **Short answer** 2 marks

Find the area of the region bounded by $y = x^3$, the x-axis and the lines $x = -2$ and $x = 1$.

Formula / idea: A part of the region below the x-axis is taken with modulus.

1 Sketch: $x^3 < 0$ for $x < 0$ and $x^3 > 0$ for $x > 0$.



2 Split at $x = 0$ and write the area. ✓ 1

$$A = \left| \int_{-2}^{0} x^3 \, dx \right| + \int_{0}^{1} x^3 \, dx$$

3 Evaluate each integral. ✓ ½

$$\int_{-2}^{0} x^3 \, dx = \left[\frac{x^4}{4} \right]_{-2}^{0} = 0 - 4 = -4$$

$$\int_{0}^{1} x^3 \, dx = \left[\frac{x^4}{4} \right]_{0}^{1} = \frac{1}{4}$$

4 Add the two areas.

$$A = |-4| + \frac{1}{4} = \frac{17}{4}$$

5 Hence the required area $= \frac{17}{4}$ sq units. ✓ ½

Answer $\frac{17}{4}$ **sq units**

Watch out: Integrating straight from -2 to 1 gives $-\frac{15}{4}$, which is wrong.

Level 3

27 marks

15 **Long answer** 5 marks

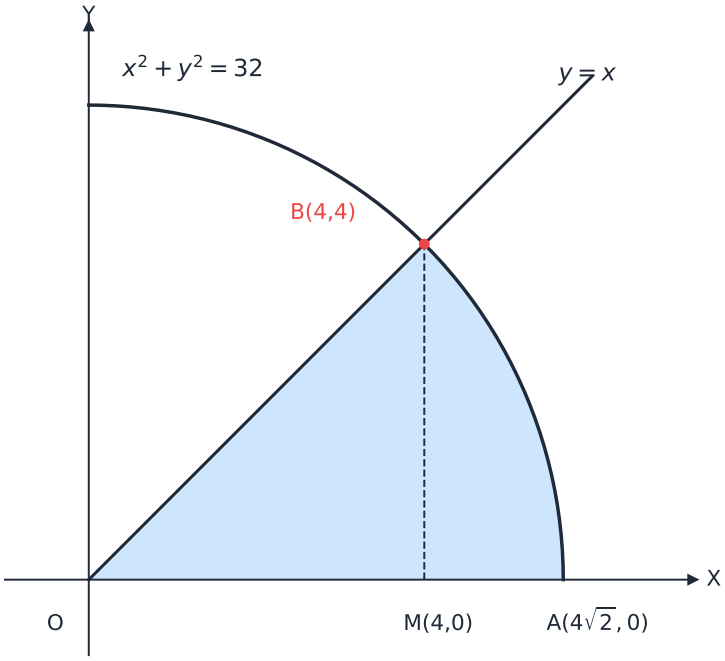
Find the area of the region in the first quadrant enclosed by the x-axis, the line $y = x$ and the circle $x^2 + y^2 = 32$.

Formula / idea: $\int \sqrt{a^2 - x^2}\,dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C$

1 Solve $y = x$ with $x^2 + y^2 = 32$.
$x^2 + x^2 = 32 \Rightarrow x^2 = 16 \Rightarrow x = 4$
Take $x > 0$ in the first quadrant.

2 So the line meets the circle at $B(4, 4)$; the circle meets the x-axis at $A(4\sqrt{2}, 0)$. ✓ ½

3 Draw the sketch and $BM \perp$ x-axis; area $=$ ar(OBM) + ar(BMA). ✓ 1



4 Write the required area. ✓ 1

$$A = \int_0^4 x\,dx + \int_4^{4\sqrt{2}} \sqrt{32 - x^2}\,dx$$

5 Area of OBM (under the line). ✓ ½

$$\int_0^4 x\,dx = \left[\frac{x^2}{2}\right]_0^4 = 8$$

6 Area of BMA: use the $\sqrt{a^2 - x^2}$ formula with $a = 4\sqrt{2}$. ✓ ½

$$\int_4^{4\sqrt{2}} \sqrt{32 - x^2}\,dx = \left[\frac{x}{2}\sqrt{32 - x^2} + 16\sin^{-1}\frac{x}{4\sqrt{2}}\right]_4^{4\sqrt{2}}$$

7 Put the limits. ✓ 1

$= \left(0 + 16 \cdot \frac{\pi}{2}\right) - \left(2 \cdot 4 + 16 \cdot \frac{\pi}{4}\right)$

$= 8\pi - 8 - 4\pi = 4\pi - 8$

$\sin^{-1} \frac{4}{4\sqrt{2}} = \sin^{-1} \frac{1}{\sqrt{2}} = \frac{\pi}{4}$

8 Add the two areas.

$A = 8 + (4\pi - 8) = 4\pi$

9 Hence the required area $= 4\pi$ sq units. ✓ ½

Answer 4π **sq units**

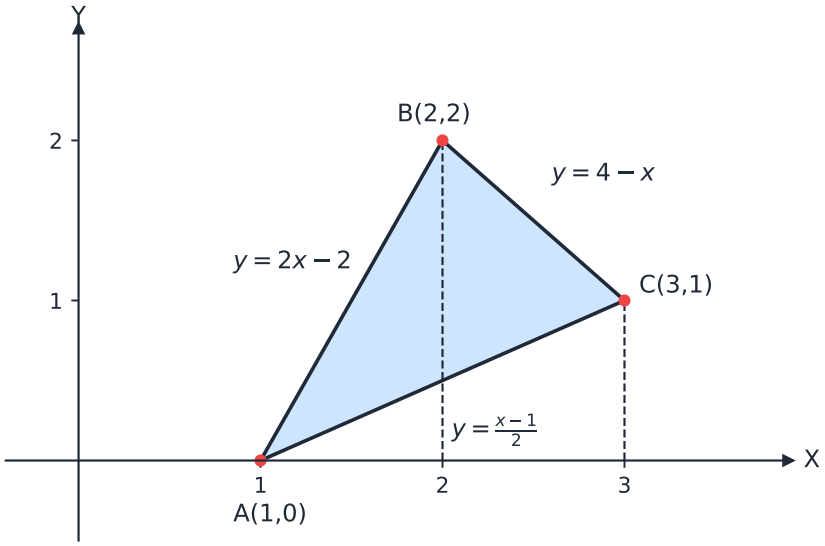
Watch out: Integrating the circle from 0 instead of from $x = 4$, where the line meets it.

16 **Long answer** 5 marks

Using integration, find the area of the triangle whose vertices are $(1, 0)$, $(2, 2)$ and $(3, 1)$.

Formula / idea: Line through two points: $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$.

1 Plot $A(1, 0)$, $B(2, 2)$, $C(3, 1)$ and shade triangle ABC. ✓ 1



2 Equation of AB. ✓ ½

$y - 0 = \frac{2 - 0}{2 - 1}(x - 1) \Rightarrow y = 2x - 2$

3 Equation of BC. ✓ ½

$y - 2 = \frac{1 - 2}{3 - 2}(x - 2) \Rightarrow y = 4 - x$

4 Equation of AC. ✓ ½

$y - 0 = \frac{1 - 0}{3 - 1}(x - 1) \Rightarrow y = \frac{x - 1}{2}$

5 Area under AB and BC, minus area under AC. ✓ 1

$$\text{ar}(ABC) = \int_1^2 (2x-2)\,dx + \int_2^3 (4-x)\,dx - \int_1^3 \frac{x-1}{2}\,dx$$

6 Evaluate the first two integrals. ✓ ½

$$\int_1^2 (2x-2)\,dx = \left[x^2 - 2x\right]_1^2 = 0 - (-1) = 1$$

$$\int_2^3 (4-x)\,dx = \left[4x - \frac{x^2}{2}\right]_2^3 = \frac{15}{2} - 6 = \frac{3}{2}$$

7 Evaluate the third integral. ✓ ½

$$\int_1^3 \frac{x-1}{2}\,dx = \frac{1}{2}\left[\frac{(x-1)^2}{2}\right]_1^3 = \frac{1}{2}\cdot 2 = 1$$

8 Combine.

$$\text{ar}(ABC) = 1 + \frac{3}{2} - 1 = \frac{3}{2}$$

9 Hence the area of the triangle $= \frac{3}{2}$ sq units. ✓ ½

Answer $\frac{3}{2}$ **sq units**

Watch out: Adding the area under AC instead of subtracting it.

17 **Long answer** 5 marks

Find the area of the region bounded by the curve $y = 4x^2$ and the line $y = 8x + 12$.

Formula / idea: Area between curves $= \int_a^b (y_{\text{upper}} - y_{\text{lower}})\,dx$.

1 Solve the two equations together.

$$4x^2 = 8x + 12 \Rightarrow x^2 - 2x - 3 = 0$$

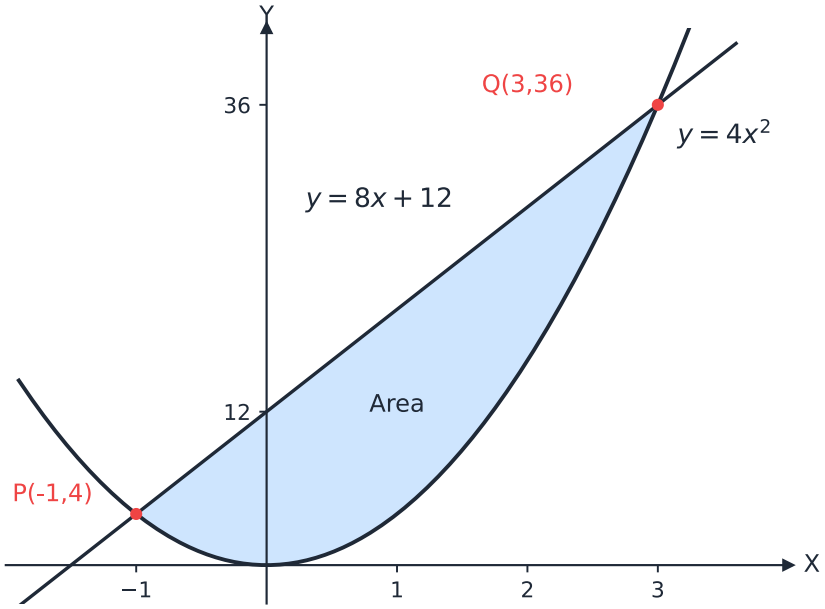
2 Factorise. ✓ ½

$$(x-3)(x+1) = 0 \Rightarrow x = -1 \text{ or } x = 3$$

3 Find the points of intersection. ✓ ½

$$P(-1, 4), \quad Q(3, 36)$$

4 Draw the sketch; between P and Q the line is above the parabola. ✓ 1



5 Write the required area. ✓ 1

$$A = \int_{-1}^{3} \left[(8x + 12) - 4x^2\right] dx$$

6 Integrate. ✓ ½

$$A = \left[4x^2 + 12x - \frac{4x^3}{3}\right]_{-1}^{3}$$

7 Put the limits. ✓ 1

$$A = (36 + 36 - 36) - \left(4 - 12 + \frac{4}{3}\right)$$
$$= 36 + \frac{20}{3} = \frac{128}{3}$$

8 Hence the required area $= \frac{128}{3}$ sq units. ✓ ½

Answer $\frac{128}{3}$ **sq units**

Watch out: Sign slip at $x = -1$: $-\frac{4(-1)^3}{3} = +\frac{4}{3}$.

18 **Long answer** 5 marks

Using integration, find the area of the region bounded by the curve $4x^2 + y^2 = 36$.

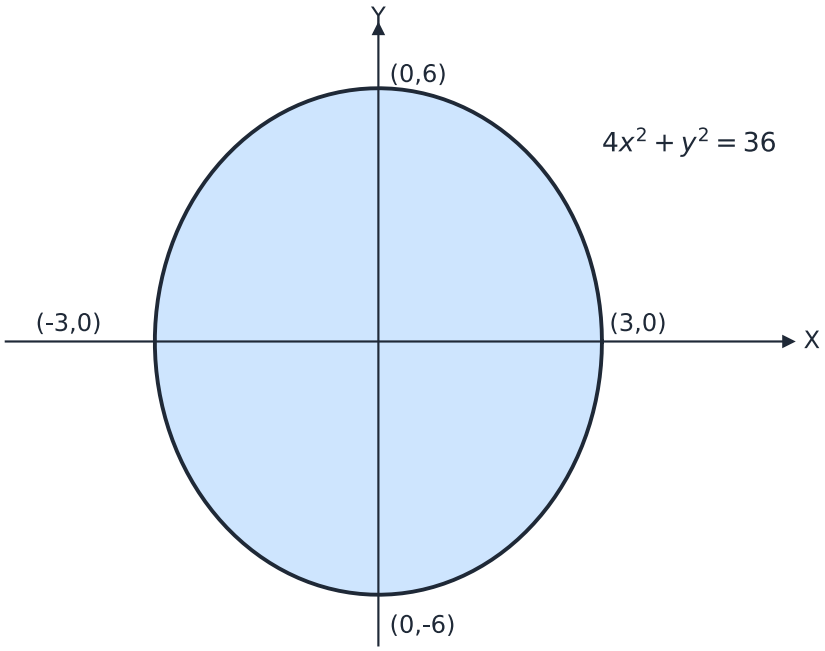
Formula / idea: $\int \sqrt{a^2 - x^2}\, dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C$

1 Divide by 36 to get the standard form. ✓ ½

$$\frac{x^2}{9} + \frac{y^2}{36} = 1$$

2 It is an ellipse with vertices $(\pm 3, 0)$ and $(0, \pm 6)$.

3 Draw the sketch and shade the whole ellipse. ✓ 1



4 Solve for y in the first quadrant. ✓ ½

$y^2 = 36 - 4x^2 \Rightarrow y = 2\sqrt{9 - x^2}$

5 Area $= 4\times$ area in the first quadrant. ✓ 1

$$A = 4\int_0^3 y\,dx = 4\int_0^3 2\sqrt{9-x^2}\,dx = 8\int_0^3 \sqrt{9-x^2}\,dx$$

The ellipse is symmetric about both axes.

6 Integrate using the $\sqrt{a^2 - x^2}$ formula with $a = 3$. ✓ ½

$$A = 8\left[\frac{x}{2}\sqrt{9-x^2} + \frac{9}{2}\sin^{-1}\frac{x}{3}\right]_0^3$$

7 Put the limits. ✓ 1

$$A = 8\left[\left(0 + \frac{9}{2}\cdot\frac{\pi}{2}\right) - 0\right] = 8\cdot\frac{9\pi}{4} = 18\pi$$

8 Hence the required area $= 18\pi$ sq units. ✓ ½

Answer 18π **sq units**

Watch out: Forgetting to multiply by 4 and giving only the quarter area $\frac{9\pi}{2}$.

19 **Short answer** 3 marks

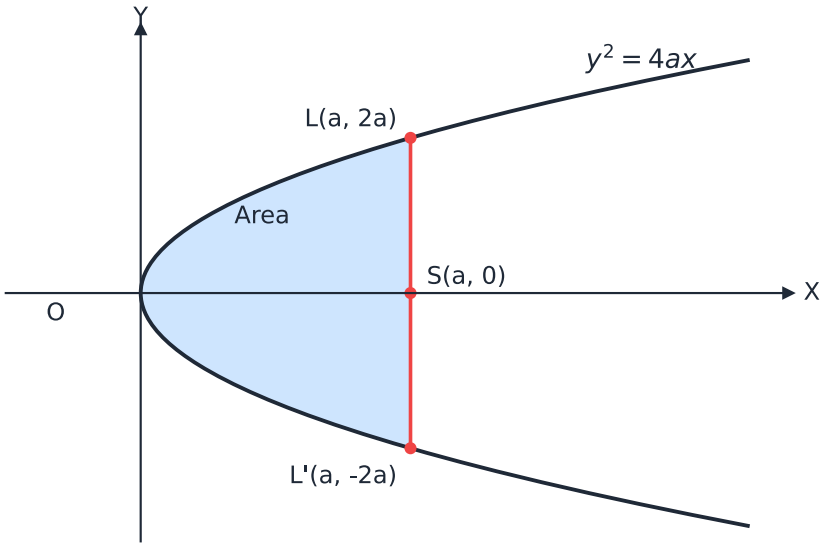
Find the area of the region bounded by the parabola $y^2 = 4ax$ and its latus rectum.

Formula / idea: Latus rectum of $y^2 = 4ax$ is the line $x = a$ through the focus $S(a, 0)$.

1 Put $x = a$ in the parabola to find the ends of the latus rectum. ✓ ½

$y^2 = 4a^2 \Rightarrow y = \pm 2a$

2 So $L(a, 2a)$ and $L'(a, -2a)$; draw the sketch. ✓ ½



3 The region is symmetric about the x-axis. ✓ 1

$$A = 2\int_0^a y\,dx = 2\int_0^a 2\sqrt{ax}\,dx$$

Area $OLL' = 2\times$ area OLS.

4 Integrate. ✓ ½

$$A = 4\sqrt{a}\left[\frac{2}{3}x^{3/2}\right]_0^a = \frac{8\sqrt{a}}{3}\cdot a^{3/2}$$

5 Simplify: $\sqrt{a}\cdot a^{3/2} = a^2$.

$$A = \frac{8a^2}{3}$$

6 Hence the required area $= \frac{8a^2}{3}$ sq units. ✓ ½

Answer $\frac{8a^2}{3}$ **sq units**

Watch out: Taking only the upper half and getting $\frac{4a^2}{3}$.

20 **Case study** 4 marks

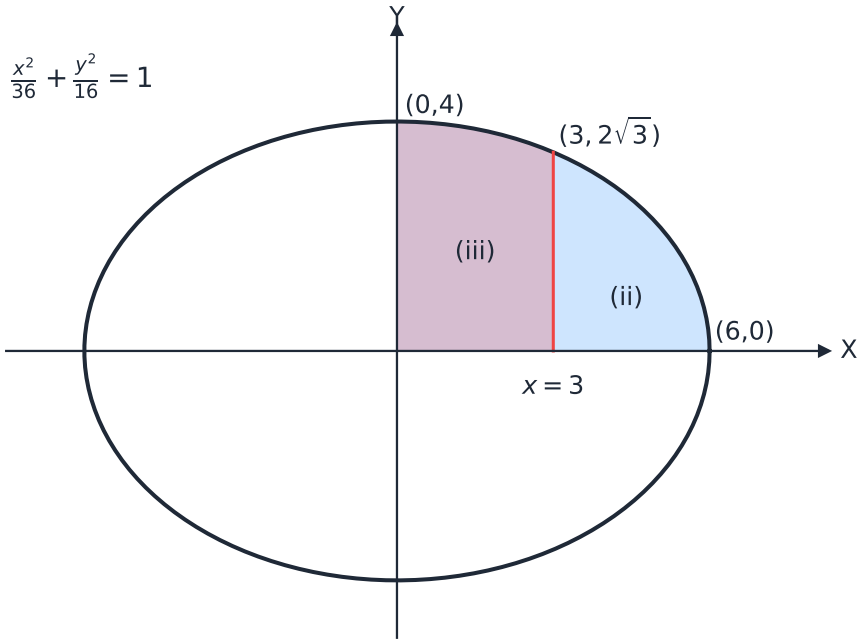
Case Study: A running track in a sports complex is in the shape of the ellipse $\frac{x^2}{36}+\frac{y^2}{16}=1$ (units in metres). The region inside the track is to be grassed.

(i) Find the total area enclosed by the track. 1 mark

(ii) Find the area of the region in the first quadrant enclosed by the track and the axes. 1 mark

(iii) Find the area of the region inside the track between $x=0$ and $x=3$ in the first quadrant. 2 marks

Formula / idea: Area of an ellipse is πab; part areas use $\int \sqrt{a^2-x^2}\,dx$.



(i) 1 mark

1 Compare with $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.
$a=6, \quad b=4$

2 Use area of ellipse $=\pi ab$. ✓ 1
$A=\pi(6)(4)=24\pi$

3 Total area enclosed $=24\pi\ \text{m}^2$.

Answer (i) $24\pi\ \text{m}^2$

(ii) 1 mark

1 The ellipse is symmetric about both axes, so take one quarter. ✓ 1
$A=\frac{1}{4}\times 24\pi=6\pi$

2 First-quadrant area $=6\pi\ \text{m}^2$.

Answer (ii) $6\pi \text{ m}^2$

(iii)

2 marks

1	Solve for y in the first quadrant. $y = \frac{4}{6}\sqrt{36 - x^2} = \frac{2}{3}\sqrt{36 - x^2}$	✓ ½
2	Write the required area. $A = \int_0^3 \frac{2}{3}\sqrt{36 - x^2}\, dx$	✓ ½
3	Integrate using the $\sqrt{a^2 - x^2}$ formula with $a = 6$. $A = \frac{2}{3}\left[\frac{x}{2}\sqrt{36 - x^2} + 18\sin^{-1}\frac{x}{6}\right]_0^3$	✓ ½
4	Put the limits. $A = \frac{2}{3}\left[\frac{3}{2}\sqrt{27} + 18 \cdot \frac{\pi}{6}\right]$ $= \frac{2}{3}\left[\frac{9\sqrt{3}}{2} + 3\pi\right]$ $\sin^{-1}\frac{1}{2} = \frac{\pi}{6}$, $\sqrt{27} = 3\sqrt{3}$	
5	Hence the required area $= (3\sqrt{3} + 2\pi)\text{ m}^2$.	✓ ½

Answer (iii) $(3\sqrt{3} + 2\pi)\text{ m}^2$

Watch out: In (iii), forgetting the factor $\frac{2}{3}$ that comes from $y = \frac{4}{6}\sqrt{36 - x^2}$.